Outset · AI in Healthcare · Day 3 · Group 3

Stroke on a brain CT — and the fairness you can't measure

A working stroke detector, and the audit this dataset makes impossible

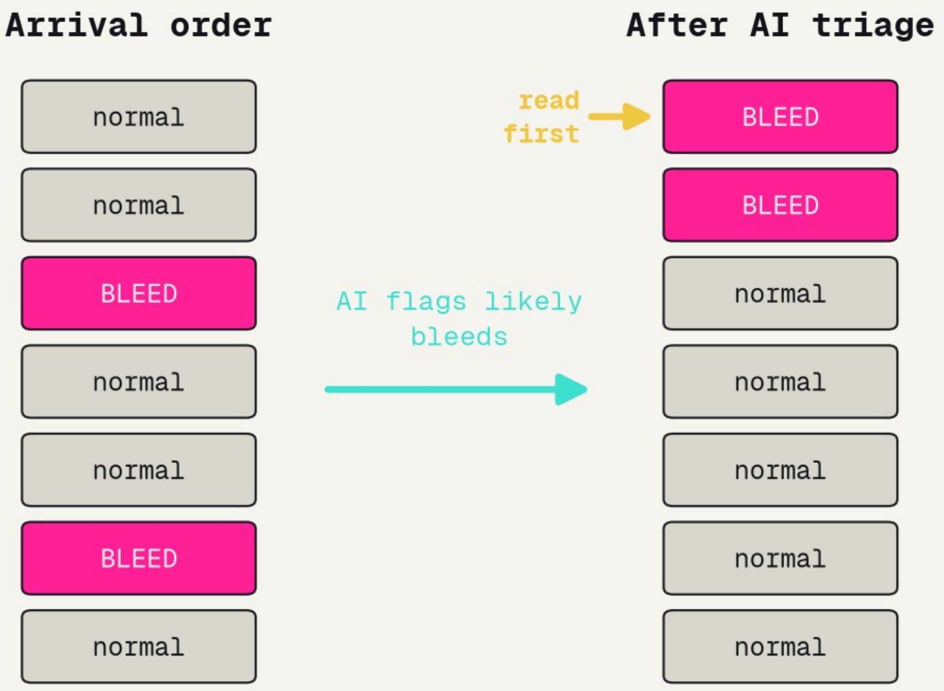
Jinchi Wei

Outset

2026-07-08

BACKGROUND

Head-CT triage AI, in one picture



Real triage tools move likely bleeds to the top of the list, cutting time-to-treatment.

adapted from Chilamkurthy et al. 2018 / Seyam et al. 2022

Triage, not diagnosis

In the ER a bleed is an emergency and minutes matter. Deployed head-CT AI -- Chilamkurthy 2018 on ~313,000 scans, the RSNA 2019 challenge, Seyam 2022 -- reorders the radiologist reading queue so likely bleeds are read first. Real tools clear ~0.90+ AUC, and still do worse on some bleed subtypes.

A dataset should tell you who is in it

What a datasheet asks for

- [x] Age
- [x] Sex
- [x] Race / ethnicity
- [x] Scanner / site
- [x] Label source
- [x] Patient consent

What our brain-CT dataset has

- [x] CT image
- [x] Stroke / normal label
- [ ] Age
- [ ] Sex
- [ ] Race / ethnicity
- [ ] Scanner / site

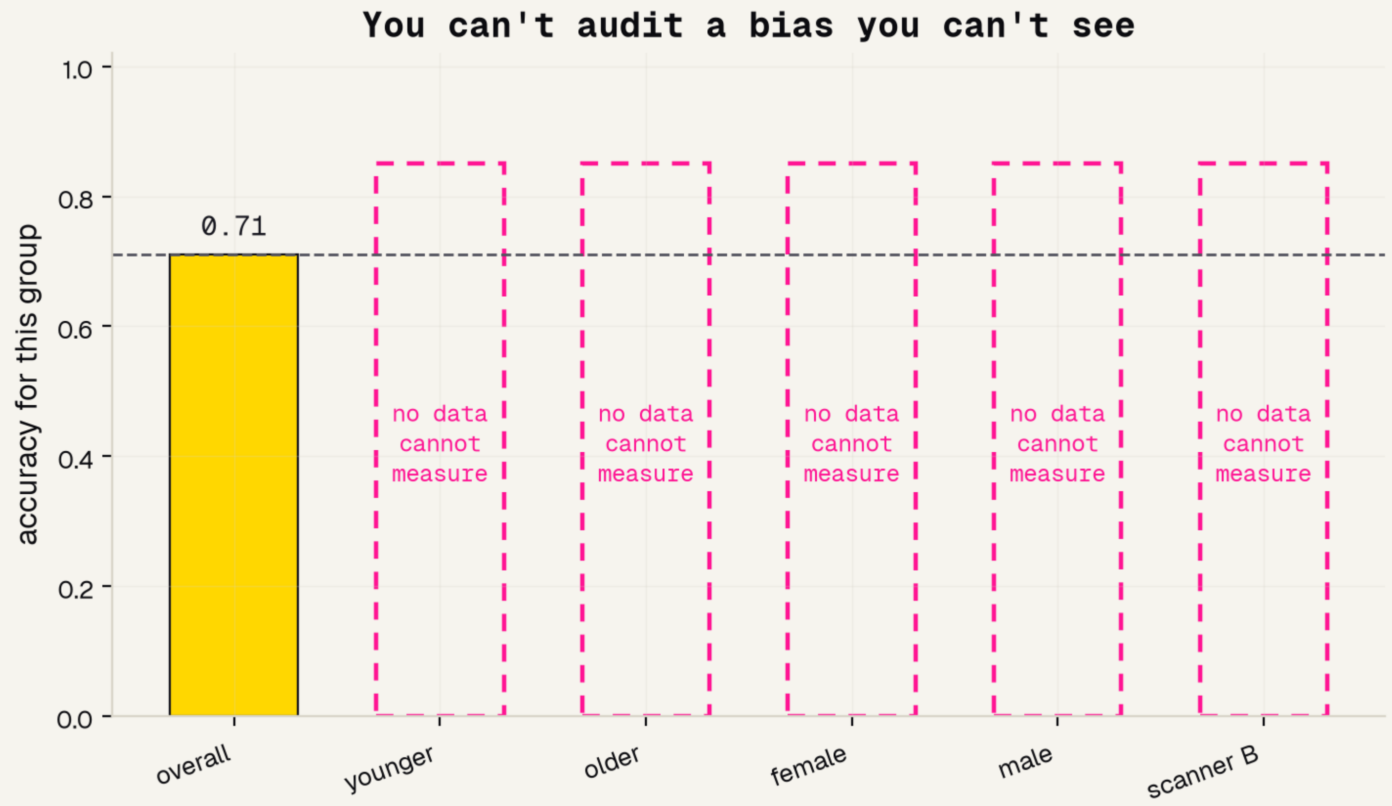
Four of the six questions have no answer -- so we cannot check who the model fails.

concept from Gebru et al. 2021 (Datasheets for Datasets) / Tripathi et al. 2023

Who is in the data?

To test fairness you compare accuracy across groups -- age, sex, race, scanner. That only works if the dataset records who each scan came from. Datasheets for Datasets (Gebru 2021) says every dataset should ship that documentation; radiology-AI reviews (Tripathi 2023) show it often does not. Ours has only the image and a label.

You can’t audit a bias you can’t see



If age / sex / race / scanner were recorded, we could fill these bars in and compare.

concept from Seyyed-Kalantari et al. 2021 / Yang et al. 2024

Blank bars

Real audits found imaging AI that missed disease more often in female, Black, and low-income patients (Seyyed-Kalantari 2021), and models that secretly lean on demographic shortcuts (Yang 2024). You can only SEE those failures if the data records the groups. When it does not, every per-group accuracy bar is a blank you cannot fill.

Methodology

METHODS

The task and the data

Task

Classify one 64x64 brain CT slice as normal or stroke. A clean binary question.

Given

The CT image, and a normal / stroke label. That is the entire dataset.

Missing

Age, sex, race, scanner, site, time-since-injury -- every field a fairness audit would need.

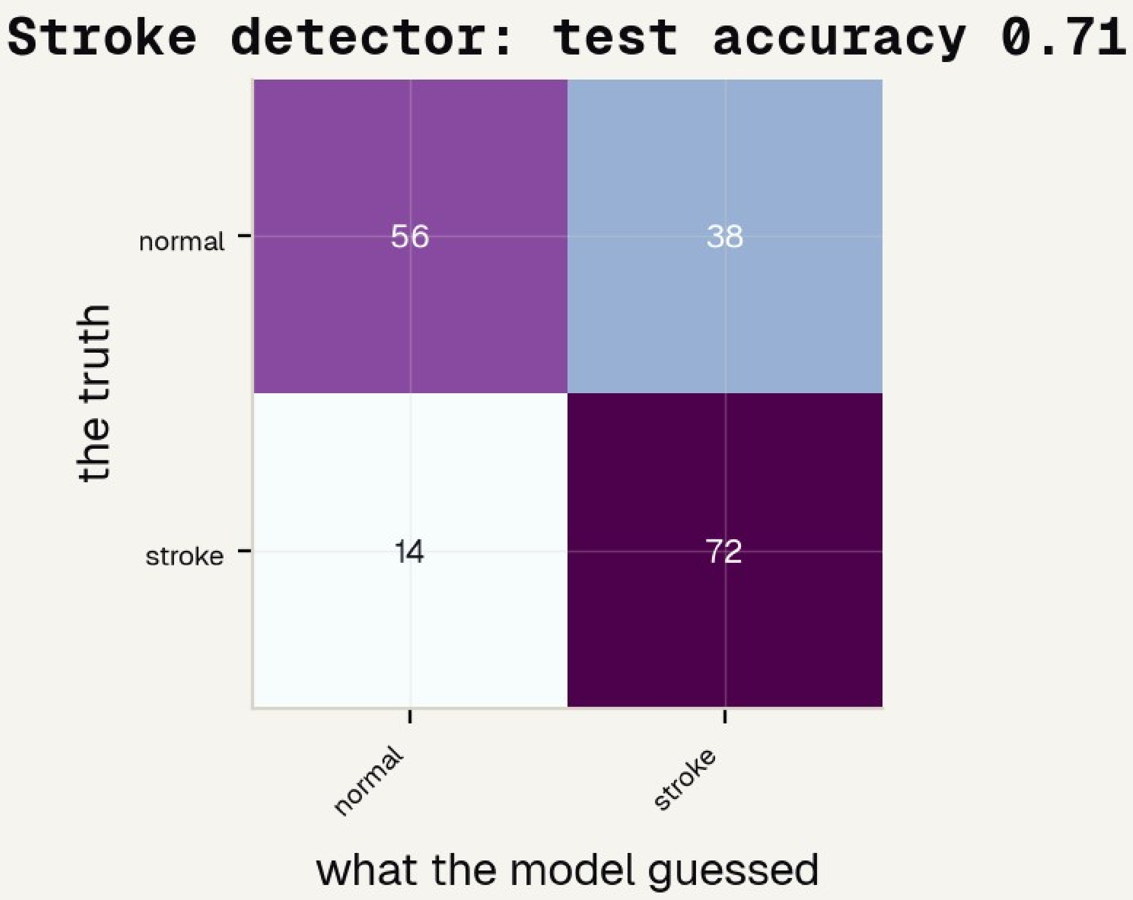
The idea: reuse a pretrained network

The same trick that won Day 1. Rather than teach a network to see from zero on a few hundred scans, we take a ResNet18 already trained on a million everyday photos, freeze what it learned about edges and texture, and train only a small new head to map those features to normal vs. stroke. Borrowed knowledge beats from-scratch when data is small.

Pretrained ResNet18 > **Freeze the body** > **Train a new 2-class head**

RESULTS

It catches most strokes — but cries wolf

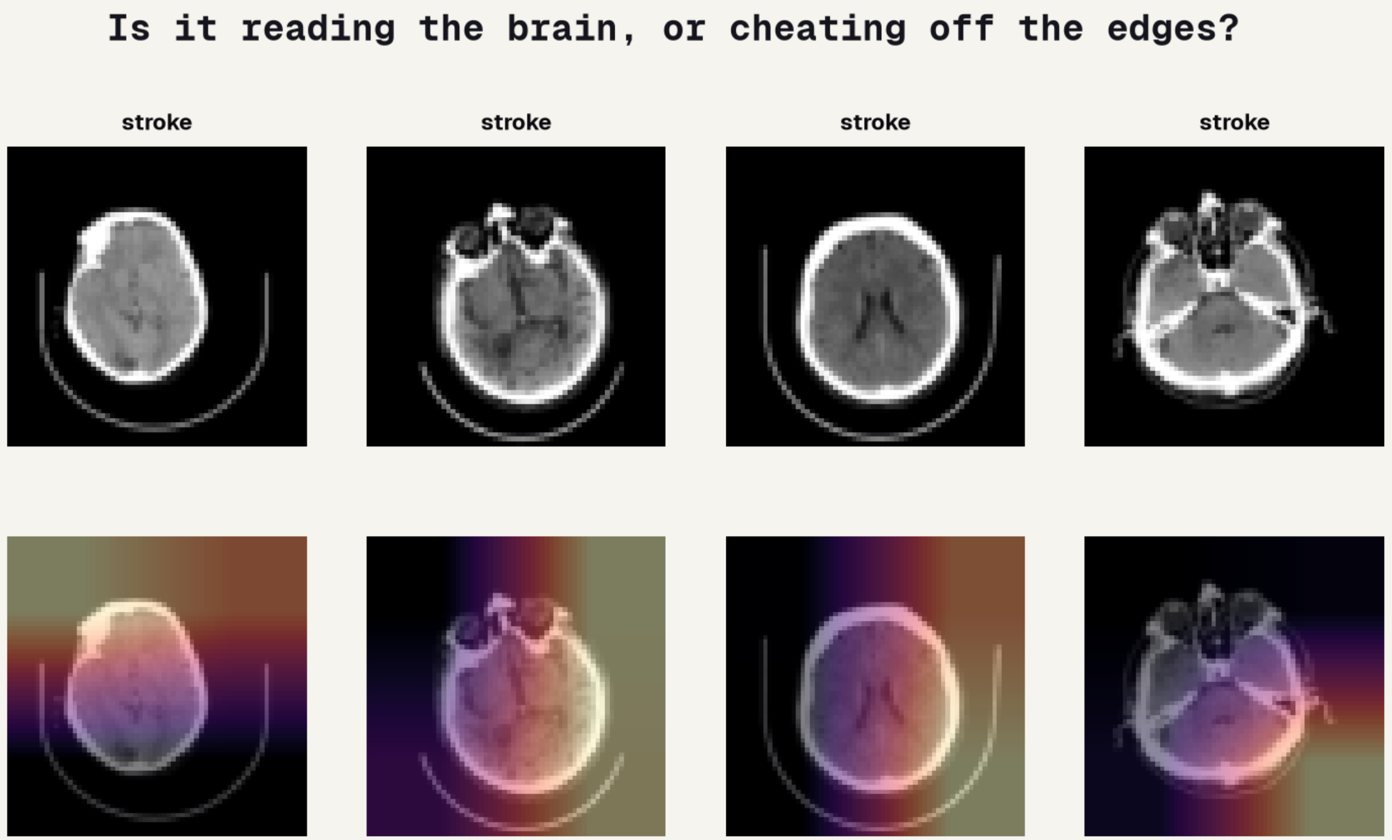


0.84

sensitivity -- of the real strokes, the share we CAUGHT

Accuracy is only ~0.71 and specificity ~0.60, so it false-alarms on many normal scans. For a safety-net triage tool, catching bleeds matters more than the false alarms -- and one accuracy number alone would have hidden that split.

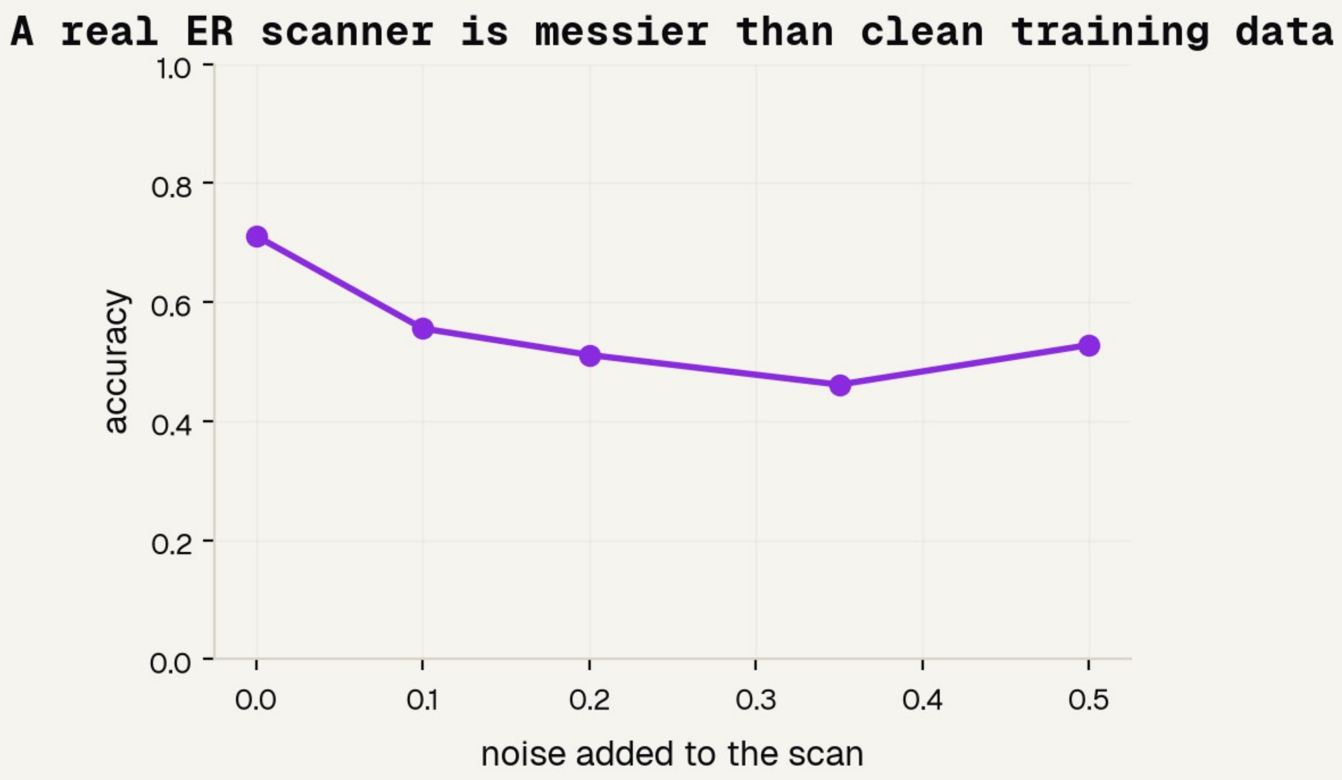
Is it reading the brain, or cheating off the skull?



Right for the wrong reason?

Grad-CAM paints WHERE the network looked. Some heat lands on brain tissue, but some drifts to the skull edge and the image border -- a classic shortcut that works on this dataset and would break on a new scanner (Yang 2024). Worth saying out loud, not hiding.

Does it survive a messier scan?



Brittle under noise

Training images are clean; a real ER scanner is noisier -- motion, low dose, different machines. Adding random noise to the test scans makes accuracy sag fast. A deployable tool would need augmentation, far more data, and live monitoring after it ships.

Validation

THE GAP THAT IS THE FINDING

The audit we literally cannot run

What this dataset tells us about each patient

GIVEN: the CT image + normal / stroke label

MISSING:
- ✗ age
- ✗ sex
- ✗ race / ethnicity
- ✗ scanner model
- ✗ hospital / site
- ✗ time since injury

You cannot audit fairness across groups the data never records.

Nothing to group by

We CAN audit fairness by class, because the label exists. But the audit that matters -- does it work as well for older patients as younger, women as men, one scanner as another -- cannot run at all. The dataset never recorded age, sex, race, or scanner. There is literally nothing to group by.

Why that silence is disqualifying

An un-runnable fairness audit is not a footnote.

Our detector might be much worse for some group of patients, and we would have no way to know. Datasheets work and radiology-AI reviews say a dataset should document who is in it; real audits show hidden bias appears exactly along age, sex, race, and scanner lines -- and only where those labels exist. For a tool that would touch patients, that silence is the headline.

REFERENCES

References

Head-CT triage AI

Chilamkurthy 2018 (The Lancet); Flanders 2020 (Radiology: AI, RSNA challenge); Seyam 2022 (Radiology: AI).

Datasets & documentation

Gebru 2021 (Communications of the ACM, Datasheets for Datasets); Tripathi 2023 (J Am Coll Radiol).

Fairness audits that need metadata

Seyyed-Kalantari 2021 (Nature Medicine); Yang 2024 (Nature Medicine).

Thank You

魏晉祺

Jinchi Wei

Outset

jinchikwei@gmail.com

夢想